# SQL ANALYTICAL PROJECT

## Business Insights from Sakila DVD Rental Database

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## 1. PROJECT OBJECTIVES

The overall objective of this project is to demonstrate how SQL queries can be used to derive business-relevant insights from a transactional relational database. Specifically, the project aims to:

- Understand **customer participation and spending** patterns across the rental business.
- Quantify **total revenue** and **rental volumes** generated by the system.
- Identify **high-demand films** and **high-revenue categories**.
- Detect **seasonal or monthly trends** in rental activity.
- Compare **store-level performance** based on revenue and rental volume.
- Analyse **actors** appearing in highly rented films to understand possible influence on demand.

## 2. TOOLS AND TECHNOLOGIES

The analysis relies on standard SQL tooling and the official Sakila sample dataset.

### 2.1 Tooling Stack

| Component | Role in the Project |
|---|---|
| MySQL Workbench | Graphical interface for query development, execution, and schema exploration. |
| SQL (Structured Query Language) | Primary language for data extraction, transformation, and aggregation. |
| Sakila Sample Database | Source dataset modelling the DVD rental business domain. |

## 3. DATASET OVERVIEW

The Sakila Sample Database simulates a DVD rental company with multiple stores, customers, staff, and a diverse movie catalog. It contains tables that record core operational entities and their relationships.

### 3.1 Core Entities

| Entity | Description |
|---|---|
| Customers | Individuals who rent movies; stored in the customer table. |
| Films | Movies available for rental, with attributes such as title, length, and rental rate. |
| Rentals | Individual rental transactions, linking customers to specific inventory copies. |
| Payments | Monetary transactions capturing revenue from rentals. |
| Inventory | Physical copies of films at specific stores. |

| Entity | Description |
|---|---|
| Stores | Store locations that manage customers, staff, inventory, and rentals. |
| Actors | Performers linked to films through a many-to-many relationship. |
| Categories | Genres grouping films (e.g., Sports, Family, Action). |

### 3.2 Data Volumes

Based on query results against the Sakila database in this project:

- **≈ 16,000 rental records**, representing the core transaction volume.
- **≈ 67,000** total revenue captured via the payment table.
- **1,000 films** in the catalog, covering multiple categories and languages.

## 4. DATABASE SCHEMA AND RELATIONSHIPS

The Sakila schema is a normalized relational model designed to support realistic rental workflows.

### Transaction and Revenue Flow

**Core rental flow:** Customer → Rental → Inventory → Film → Category

**Revenue flow:** Customer → Rental → Payment

**Actor linkage:** Actor → Film_Actor → Film

**Store operations:** Store → Staff → Payment

Store → Inventory → Rental

## 5. CUSTOMER ANALYSIS

### 5.1 Objective

To profile customer participation and spending patterns, identify high-value customers, and understand revenue concentration.

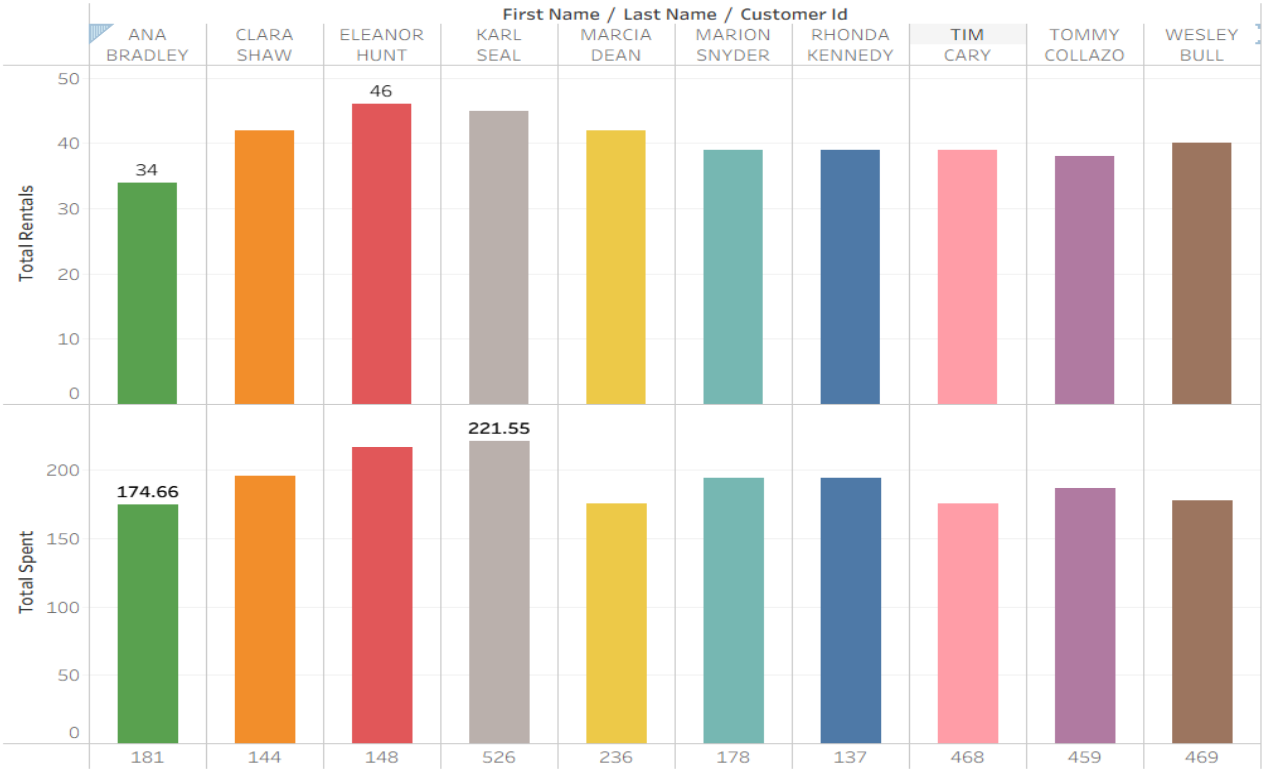
### 5.2 Key Metrics

| Metric | Result |
|---|---|
| Total Customers | 599 |
| Highest-Spending Customer | Karl Seal |
| Average amount spent by customers | 112.53180 |

### 5.3 Insights

- The system maintains **599 active customers**, providing a substantial base of repeat rental activity.
- **Karl Seal** emerges as the **highest-value customer** spending 221.55 dollars, indicating a candidate for loyalty or VIP programs.
- Identifying high-value customers enables businesses to create **loyalty programs and targeted promotions**.

### Top 10 Customers




## 6. REVENUE ANALYSIS

### 6.1 Objective

To quantify the financial performance of the rental business by measuring total revenue and transaction volume.

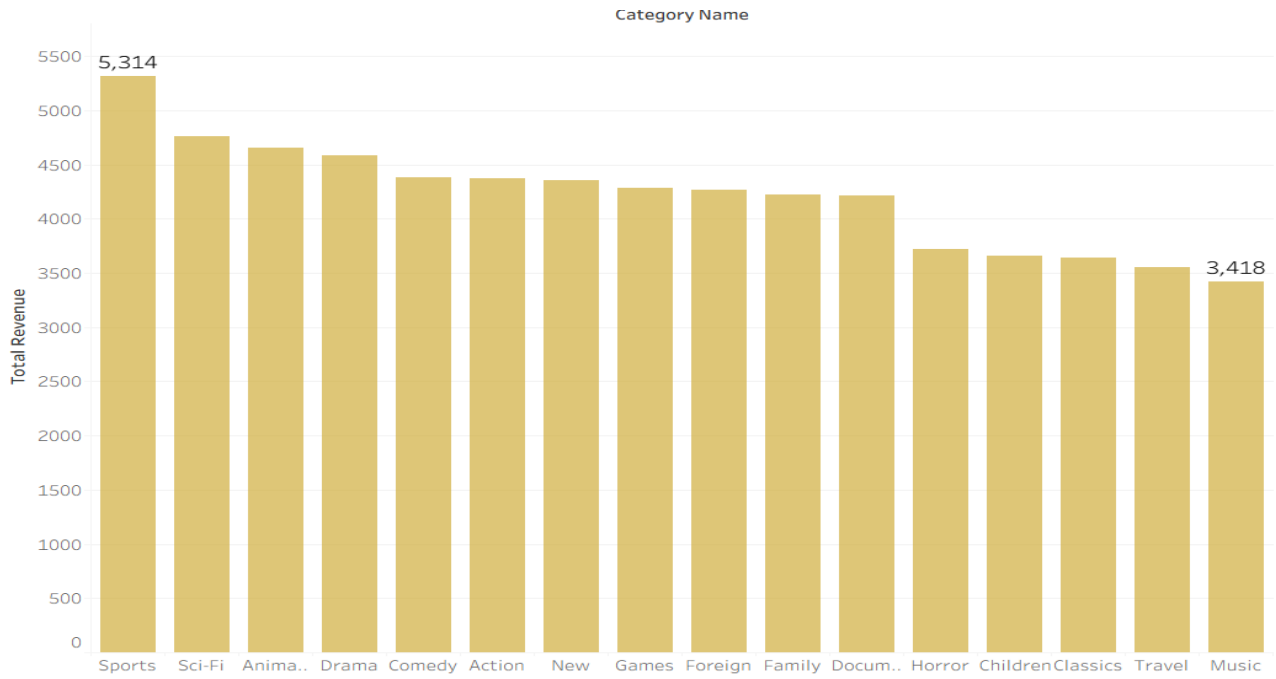
### 6.2 Key Metrics

| Metric | Result |
|---|---|
| Total Rental Transactions | ≈ 16,000 |
| Total Revenue | ≈ 67,000 |

### 6.3 Insights

- The ≈ **16,000 rentals** reflect strong customer engagement and ongoing use of the service.
- Revenue concentration analysis shows that a limited number of films and categories contribute disproportionately to total revenue.
- The **Sports** category generates the highest total rental revenue among all film categories.

## Popular Categories Collecting Highest Revenue




# 7. MOVIE AND CATEGORY ANALYSIS

## 7.1 Objective

To identify the most popular films and the categories that generate the highest revenue, supporting inventory and marketing decisions.

## 7.2 Key Metrics

| Metric | Result |
|---|---|
| Total Films | 1,000 |
| Most Rented Movie | Bucket Brotherhood |
| Top Revenue Category | Sports |

The most rented film is computed by counting rentals per film, and the top category is identified by summing payment amounts through joins between payment, rental, inventory, film, and category tables.

## 7.3 Film Popularity Insights

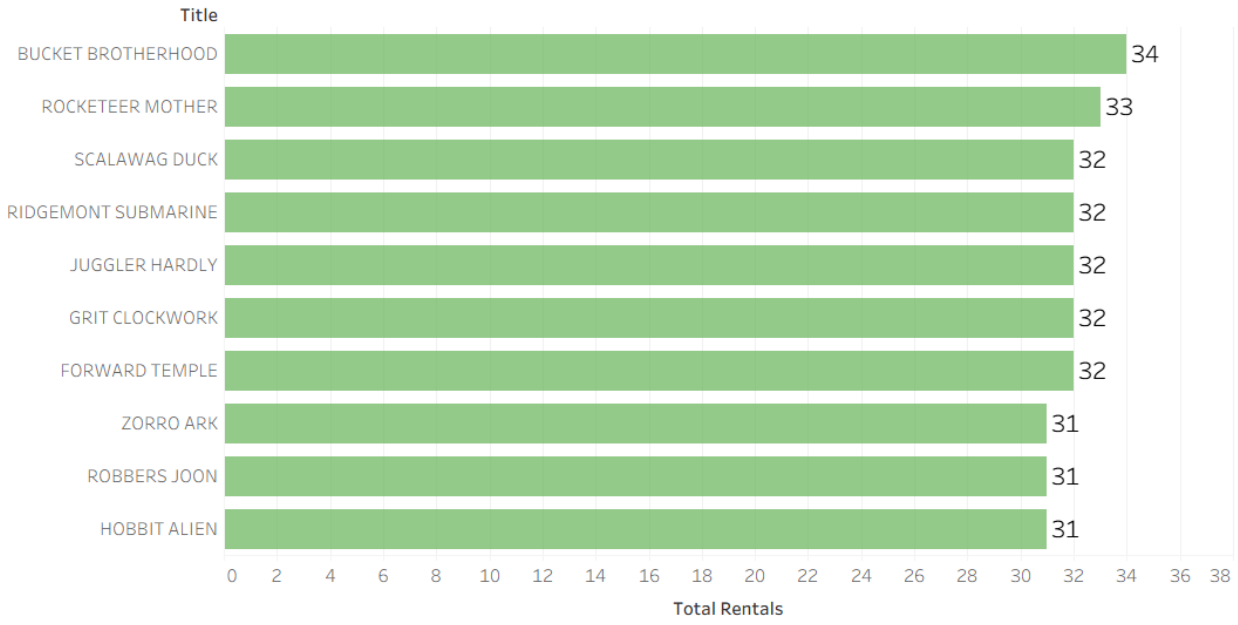
- With **1,000 titles**, the catalog offers diverse options across multiple genres and languages.
- **"Bucket Brotherhood"** has the **highest rental count**, making it a flagship title for the business.
- High-demand films warrant **higher inventory allocation** and **priority placement** in promotions and recommendations.

## 7.4 Category Revenue Insights

- The **Sports** category generates the **highest total revenue**, outperforming other genres.

- This suggests a strong **customer preference for sports-related content**, which could shape future acquisition or licensing strategies.
- Other categories can be benchmarked against Sports to evaluate under- or over-performance.

### Top 10 Most Popular Movies by Rentals




## 8. Time-Based (Seasonal) Analysis

### 8.1 Objective

To analyse rental behaviour over time and detect seasonal or monthly patterns that can inform operations and marketing.

### 8.2 Key Metrics

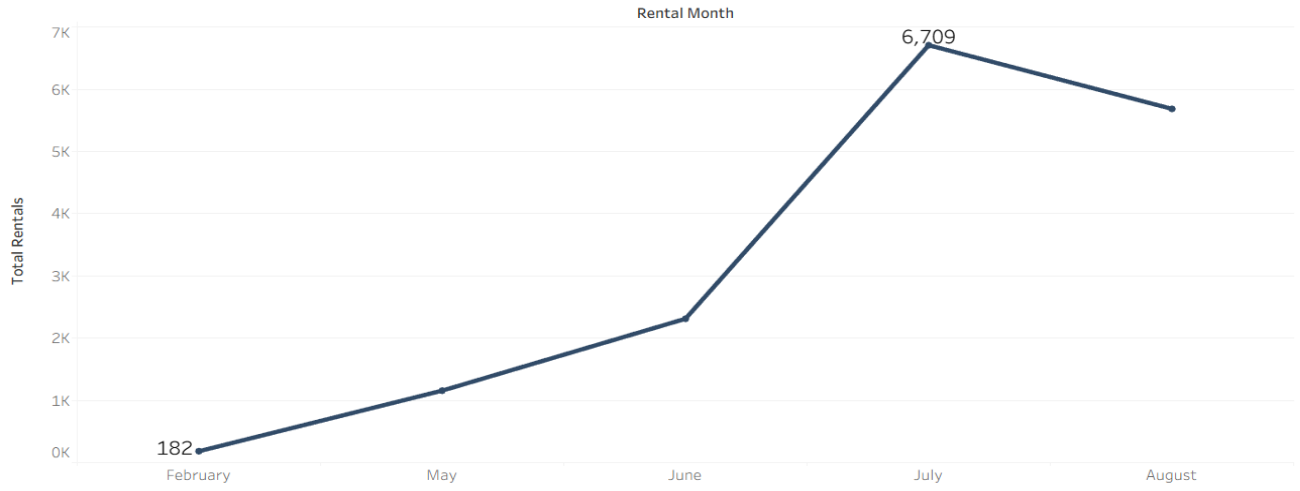
| Metric | Result |
|---|---|
| Total Rentals | ≈ 16,000 |
| Peak Month | July |

The peak month is derived by grouping rentals or payments by the month of rental_date or payment_date and selecting the month with the highest count.

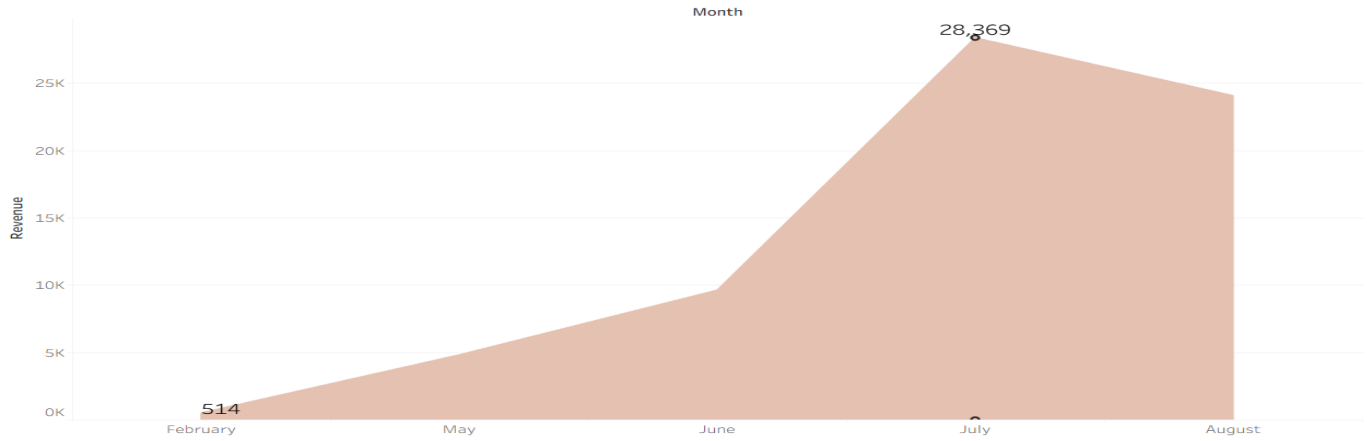
### 8.3 Insights

- Rental demand **varies across months**, indicating clear seasonality rather than evenly distributed activity.
- **July (Month 7)** records the highest rental volume, indicating a clear seasonal peak.
- Understanding temporal peaks allows the business to **plan staffing, inventory allocation, and marketing campaigns** more effectively.

**MONTHLY RENTALS**




**MONTHLY REVENUE**




# 9. STORE PERFORMANCE ANALYSIS

## 9.1 Objective

To compare the performance of the two physical store locations in terms of revenue and rental activity.

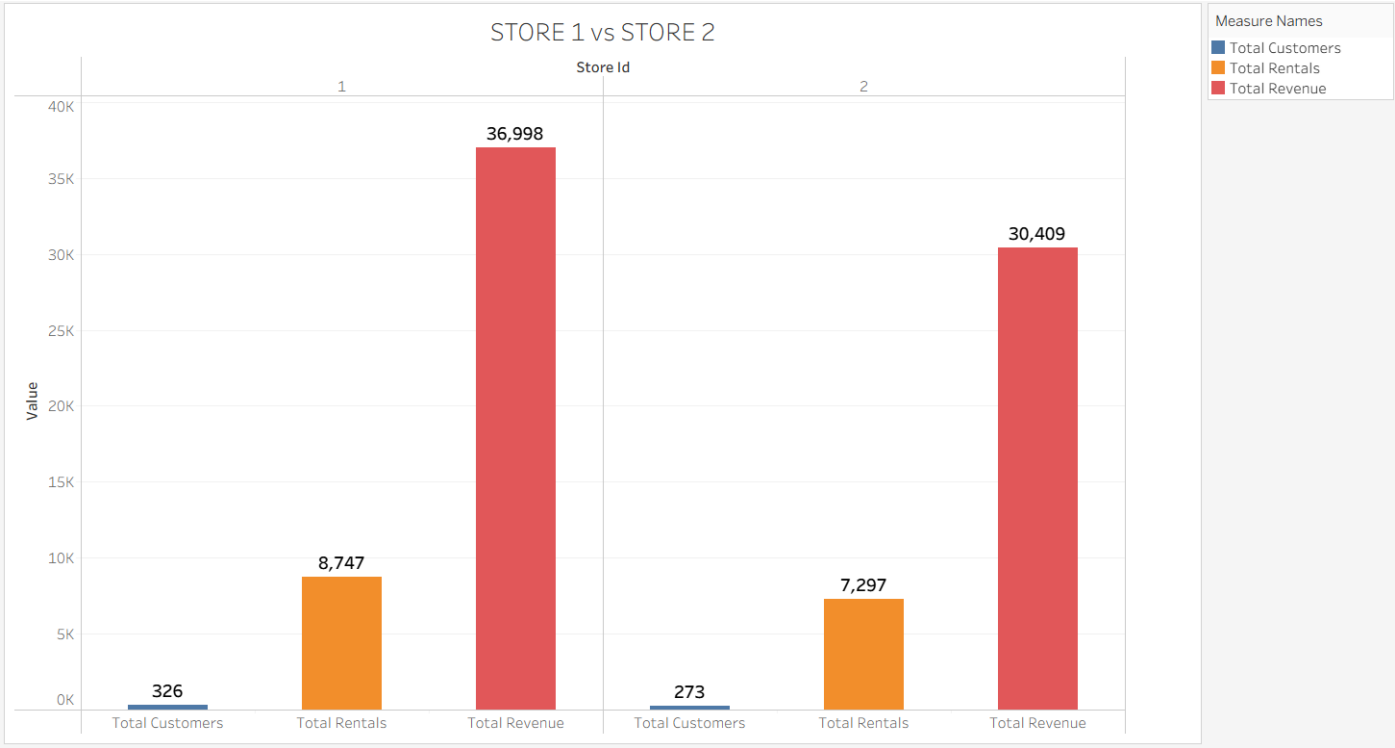
## 9.2 Key Metrics

| Metric | Result |
|---|---|
| Number of Stores | 2 |
| Higher Revenue Store | Store 2 |

Store-level revenue is computed by joining payment records to staff, then to store, and aggregating by store_id (or by using the sales_by_store view where available).

### 9.3 Insights

- The dataset includes **two stores**, both contributing meaningfully to overall revenue.
- **Store 2** generates **slightly higher revenue** than Store 1, implying either a larger customer base, higher utilization, or more effective operations.
- Differences in performance may be due to **location, customer demographics, or staff efficiency**, though these factors are not explicitly modelled in the dataset.




## 10. Actor Influence

To identify actors whose movies are rented most frequently, and assess whether actor presence may influence customer rental choices.

### Approach

- Join **actor**, **film_actor**, **film**, **inventory**, **rental**, and optionally **payment** tables.
- Count rentals per actor across all films in which they appear.
- Rank actors by total rental count (and optionally by revenue).

### Key Result

- A small group of actors appear in **highly rented films**, indicating that certain names may be associated with more popular content.
- Dina Degeneres appears in 42 films and is associated with the highest cumulative rental count, suggesting that actors with higher film participation tend to generate greater overall rental demand.

### Top 3 Actors by Film Rentals

| Rank | Actor Name | Number of Films | Total Rentals |
|---|---|---|---|
| 1 | DINA DEGENERS | 42 | 753 |

| Rank | Actor Name | Number of Films | Total Rentals |
|---|---|---|---|
| 2 | METTHEW CARREY | 37 | 678 |
| 3 | MARY KEITEL | 40 | 674 |

### Insights

- Actors who consistently appear in **high-demand films** can be leveraged in **marketing messaging** (e.g., "Featuring popular actor X").
- Films featuring these actors might warrant **additional copies** in inventory and **prominent placement** in recommendations.

## 11. CONSOLIDATED BUSINESS INSIGHTS

Bringing together the analyses across customers, revenue, films, categories, time, and stores, several overarching patterns emerge:

- The business exhibits **strong customer engagement**, with **over 16,000 rentals** across **599 customers**.
- **Total revenue of ≈ 67,000** underscores the commercial viability of the rental operation in the simulated environment.
- **"Bucket Brotherhood"** is the **most popular movie**, and the **Sports** category produces the **highest revenue**, signalling clear content preferences.
- **Karl Seal** is the **highest-value customer**, illustrating the presence of VIP-type customers whose retention is particularly important.
- The **Sports** category is the **top-revenue-generating genre**, out-performing other categories.
- **Month 7** (July) exhibits the **peak rental activity**, pointing to clear seasonality in demand.
- **Store 2** generates slightly higher total revenue than **Store 1**, suggesting store-level performance differences.
- A small group of **actors** appear repeatedly in highly rented movies, indicating a potential link between actor presence and film demand.

These insights demonstrate how **SQL-based analysis** can uncover integrated business perspectives, product demand, revenue performance.

## 12. LIMITATIONS OF THE ANALYSIS

While the analysis produces meaningful patterns, several limitations must be acknowledged:

- The **Sakila database is a synthetic sample dataset**, designed for demonstration and training rather than representing a real company.
- Important contextual variables are **absent**, including:
    - Customer demographics (age, income, occupation).
    - Marketing campaign data and promotional history.
    - Geographic attributes beyond addresses (e.g., city-level demand analyses).
    - Competitor activity and pricing.

- The dataset is **static** and does not incorporate streaming real-time updates or long-term historical changes.
- As a result, the insights should be treated as **analytical examples and methodological templates**, not direct predictions of real-world performance.

## 13. CONCLUSION

This project demonstrates how SQL can be used to analyse a relational transactional database and derive **actionable business insights**. Leveraging the Sakila Sample Database, the analysis explored customer behaviour, movie demand, revenue patterns, seasonality, and store performance.

- Customer engagement is strong with over 16,000 rental transactions.
- A small group of customers contributes disproportionately to revenue.
- The Sports category dominates revenue generation.
- Rental demand exhibits clear seasonality with a July peak.
- Store-level performance differences suggest potential operational or demographic variations.